



Ion implantation-driven high-quality graphene growth on insulating substrates with catalyst spatial confinement

Yang Xu^{a, *}, Yue Li^{a, b}, Liu Qian^b, Shurui Wang^b, Yunbiao Zhao^{a, *}, Ziqiang Zhao^{a, b, **, *}

^a State Key Laboratory of Nuclear Physics and Technology, School of Physics, Peking University, Beijing, 100871, China

^b School of Materials Science and Engineering, Peking University, Beijing, 100871, China

ARTICLE INFO

Keywords:

Ion implantation

Catalyst spatial confinement

Insulating substrates

Graphene preparation

ABSTRACT

Graphene is widely recognized for its exceptional electrical and thermal conductivity, as well as its remarkable optical and mechanical properties. These unique characteristics make it highly attractive for applications in integrated circuits, field-effect transistors, optoelectronic devices, and sensors. However, transferring graphene grown on metal substrates to semiconductor or insulating substrates typically involves a complex procedure that may lead to material degradation and wrinkle formation. Therefore, direct growth of high-quality graphene on these substrates is particularly important. In this study, we present a synergistic strategy that combines ion implantation of metal ions with catalyst spatial confinement to grow high-quality graphene. This method not only significantly reduced the growth time and improved the uniformity of the graphene but also effectively reduced metal residues. Furthermore, we investigated the molecular behavior within micrometer-sized confined spaces, which extended the active lifespan of the catalysts and increased the supersaturation of carbon species. This synergistic strategy provides valuable guidance for the controlled growth of other two-dimensional materials.

1. Introduction

Graphene is a two-dimensional material composed of a single atomic layer arranged in a hexagonal honeycomb lattice structure [1]. Its sp^2 hybridization endows it with very excellent electrical properties [2–5], leading to its wide applications in the field of electronic devices [6,7]. The applications of graphene-based electronic devices are primarily on semiconductor or insulating substrates [8–11] rather than on metal substrates which are beneficial for graphene growth [1,12,13], and the methods for growing graphene on these substrates include the transfer method and direct growth method. The transfer method may introduce defects, wrinkles, and other issues, which can adversely affect the excellent electrical and mechanical properties of graphene [14].

Thus, directly growing graphene on the target substrate presents greater application prospects [15,16]. Direct growth of graphene on insulating substrates is very challenging due to issues such as catalytic inactivity and uneven nucleation [16]. Many methods have been explored to address the catalytic inactivity of insulating substrates, including remote metal catalysts [17,18] and spin-coating metal catalysts [8,19,20]. For example, high-quality, defect-free single-layer

graphene (SLG) was directly synthesized on dielectric substrates via a Cu-vapor-assisted chemical vapor deposition (CVD) process. However, the methane-to-hydrogen ratio must be strictly controlled, and the adsorption and deposition of copper vapor pose significant challenges [21]; Similarly, when Ge is deposited on insulating substrates such as quartz glass or Si/SiO₂, the resulting graphene typically exhibits a higher I_D/I_G ratio and requires growth times exceeding 300 min [22]. While these methods alleviate the problem of catalyst inactivity, uneven distribution of the catalysts can result in inconsistent growth and may introduce metal impurities due to potential metal residues left during the etching away of the catalysts, which is detrimental to the application of graphene.

In previous work, we demonstrated the advantages of localized catalysis by nanoparticles [23]. However, the spin-coating process resulted in non-uniform catalyst distribution, whereas the uniform catalyst loading afforded by the ion implantation technique is a significant advantage, providing more uniform graphene nucleation sites. Ion implantation is an effective technique for uniformly loading metal and non-metal ions into a substrate [9,24,25], allowing precise control over the concentration and distribution of catalysts. While this method

* Corresponding author.

** Corresponding author. State Key Laboratory of Nuclear Physics and Technology, School of Physics, Peking University, Beijing, 100871, China.

E-mail addresses: yunbiaozhao@pku.edu.cn (Y. Zhao), zqzhao@pku.edu.cn (Z. Zhao).

<https://doi.org/10.1016/j.carbon.2025.120183>

Received 2 December 2024; Received in revised form 8 February 2025; Accepted 3 March 2025

Available online 3 March 2025

0008-6223/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

showed promise in initial studies, experiments revealed that catalysts introduced via ion implantation rapidly escaped at high temperatures, with their effectiveness lasting only around 20% of the total growth time. These findings indicated significant room for improvement, particularly in extending the active lifespan of the catalysts and enhancing their overall performance during graphene growth.

In this work, we focused on extending the lifetime of catalysts loaded via ion implantation by adopting a space confinement strategy [26–29]. During the growth process, we trap the released ions at the catalytic front, allowing them to continuously function throughout the entire growth. Subsequent experimental results indicated that this system significantly improved the active time of the catalyst ions, achieving up to over 75% of the total growth time. Additionally, we used simulation methods to verify the significance of confinement in extending the active time of the catalyst ions. This approach greatly enhanced catalytic efficiency, reducing the growth time by 250% (2 h for full coverage in this work while 5 h in previous work) [9], resulting in high-quality graphene glass.

2. Experimental section

2.1. Catalysts preparation

In this experiment, the substrates used were quartz, sapphire, and Si/SiO₂. The catalyst ions (Fe, Cu, Co) were loaded into the target substrates through ion implantation by ion implanter (LKBT-400), with energies ranging from 1 to 15 keV and doses from 1×10^{14} to 1×10^{16} ions/cm². After implantation, the substrates required post-treatment for repair and to assist the migration of the catalysts to the near-surface region, which were ultrasonically cleaned in deionized water, acetone, and ethanol for 30 min in sequence before being placed in a muffle furnace (BEST EQUIPMENT, 1400 °C 1.5L MF-1400C-S) for annealing. For quartz and Si/SiO₂, we performed annealing at 900 °C for 5–9 h, while for the sapphire substrates, we conducted annealing at 1100 °C for the same duration.

2.2. Graphene growth

Graphene films were grown on the target substrates using atmospheric pressure chemical vapor deposition (APCVD). The substrates were placed in a commercially available tube furnace (BEST EQUIPMENT, BTF-1200C) with a 2 inches-diameter tube for growth. During the growth process, four growth modes were employed, including ion implantation with a gap (catalyst spatial confinement, abbreviated as ION + CSC), ion implantation without a gap (without catalyst spatial confinement, abbreviated as ION ONLY), no ion implantation with a gap, and no ion implantation without a gap. Before growth, the tube furnace was pumped below 0.5 Pa to remove air. During the growth process, 200 sccm of Ar, 30–50 sccm of H₂, and 2–5 sccm of CH₄ were introduced, with a growth temperature of 1100 °C and a growth time controlled between 10 min and 120 min. After growth, the samples undergone post-treatment in an Ar/H₂ atmosphere for 20 min, and then they were cooled by sliding the heated zone away.

2.3. Characterization of catalysts and graphene

Raman spectroscopy (Thermo DXRxi, with laser excitation at 532 nm, and power of 9.0 mW), scanning electron microscopy (SEM, measured on Hitachi S-4800 at 1.0 kV) and transmission electron microscope (TEM, FET Tecnai F20, operating at 200 kV), were used to characterize the quality and coverage area of the grown graphene, while X-ray photoelectron spectroscopy (XPS, AXIS Ultra photoelectron spectrometer using monochromatic Al K α radiation) and atomic force microscopy (AFM, obtained using a Dimension Icon microscope, Bruker) were employed for pseudo-in situ characterization of the active time of the catalyst nanoparticles and the process of their precipitation from the

substrate. Time of flight secondary ion mass spectrometry (TOF-SIMS) measurements were conducted with a TOF-SIMS IONTOF M6 instrument (IONTOF GmbH). A Bi₃⁺ (a small cluster of three Bi atoms with two charges) beam (30 keV, 0.43 pA) was used as the primary beam to detect the samples, and the sputter etching was performed using an O₂ Ion Gun (1 keV) to obtain the desired depth profile. The area of analysis was $15 \times 15 \mu\text{m}^2$, while the sputtering area was $450 \times 450 \mu\text{m}^2$.

2.4. Simulation and calculation

SRIM (Stopping and Range of Ions in Matter) was used to simulate the distribution of ions in the substrate during the ion implantation process at different implantation energies, with the implanted ion species being Fe and the implantation energies set at 5 keV, 10 keV, and 15 keV. The finite element method was employed to simulate the airflow patterns in spatially confined and unconfined systems [30,31], where the heights of the confined gap were 10 μm , 100 μm , and 1000 μm , and the simulations were conducted in a laminar flow mode (details in supplementary materials). The diffusion equation (1) was used to calculate the trend of molecular variation with respect to the height of the confined gap and time, as well as the collision probability (2) of molecules within the gap (details in the following text) [32].

3. Results and discussion

Fig. 1a schematically illustrates the growth strategy designed for the catalyst spatial confinement method, highlighting the beneficial role of the gap, and the distance of the gap was about $\sim 10 \mu\text{m}$ (shown in Fig. S1). Two series of samples of the same size were grown using different methods. In both methods, metal ions were implanted into the substrate with a fluence from 1×10^{14} to 1×10^{16} ions/cm² and energy from 1 to 15 keV. During the annealing process prior to growth, the ions migrated to the near-surface region. The key difference between the two methods lies in the growth process: specifically, whether an additional layer of quartz glass was placed on top of the substrate to create a confined space. During the growth process, Fe atoms diffused from the substrate and migrated across the surface, leading to the formation of nanoparticle clusters. In addition to surface migration, Fe atoms could also easily escape at such a high temperature from the surface. The purpose of this upper substrate was to confine the catalyst nanoparticles within the gap, allowing them to remain in place to function for a longer duration during the graphene growth process, which enhanced the cracking efficiency of the precursors. The confined space of the growth system is placed in Fig. 1b, and the length of the heating belt was about 9 cm, making it a thermostatic atmosphere. Furthermore, the different growth results are shown in Fig. 1c that the spatial confined pattern shows a lower D band and more uniform layers with Fe ions on quartz and others on Si/SiO₂ and sapphire with different ions are shown in Fig. S4, and there is a synergistic role combining ion implantation and space confinement which is also shown in Fig. 1c.

To evaluate the quality of our growth samples, we utilized Raman characterization. The Raman characteristic bands of graphene include the D band ($\sim 1350 \text{ cm}^{-1}$), G band ($\sim 1580 \text{ cm}^{-1}$), and 2D band ($\sim 2700 \text{ cm}^{-1}$) [33]. The ratio of I_D/I_G serves as an indicator of the defect density, while the ratio of I_{2D}/I_G provides an estimate of the number of graphene layers. Additionally, the signal-to-noise ratio (SNR) reflects the relative content of graphene in the measured spectral region. Fig. 2a shows the characterization results of the sample with ion implantation and a gap. As the growth time increases, the value of I_D/I_G clearly decreases. At 30 min into the growth process, the D band, which represents the level of defects, is significantly elevated. This increased D band can be attributed to the initial growth phase, during which gaseous precursors decomposed on the substrate to form nuclei, resulting in a higher number of defects. In the subsequent growth stages, carbon species were more likely to adsorb onto the edges of the nucleation sites and assemble into larger grains, leading to a substantial reduction in the D band. When the

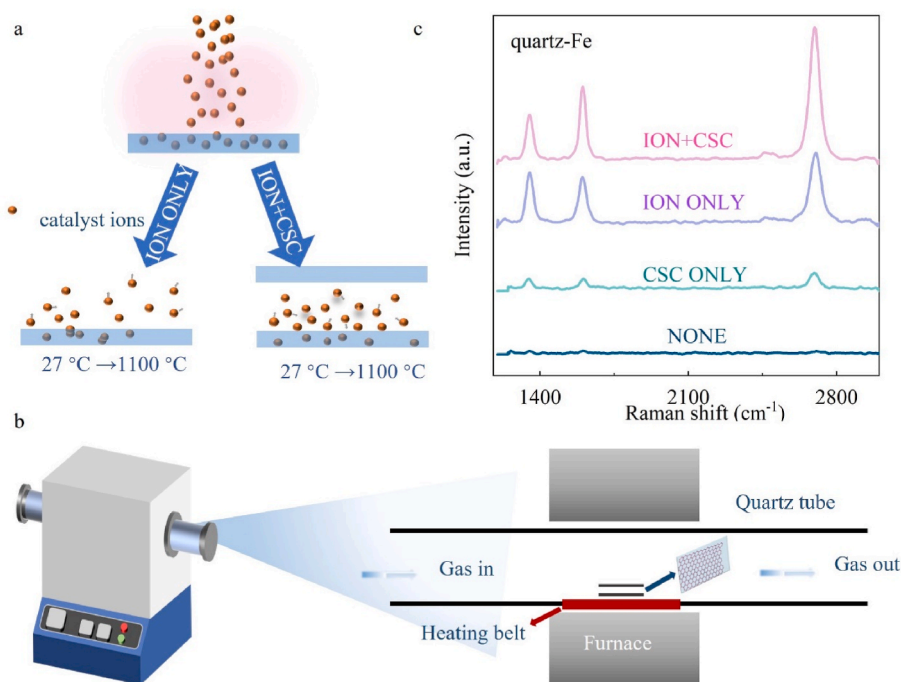


Fig. 1. (a) Schematic diagram of the catalyst ions assisted APCVD process for the synthesis of graphene on quartz glass with ION + CSC and ION ONLY patterns growth by APCVD at 1100 °C by the same flow rate of 2 sccm CH₄, 40 sccm H₂, 110 sccm Ar. (b) The whole schematic diagram of the synthesis of graphene on substrates in the APCVD system. (c) Raman spectra of graphene on quartz with the methods of ION + CSC, CSC ONLY, ION ONLY and NONE by APCVD with Fe ions at 1100 °C for 120 min. (A colour version of this figure can be viewed online.)

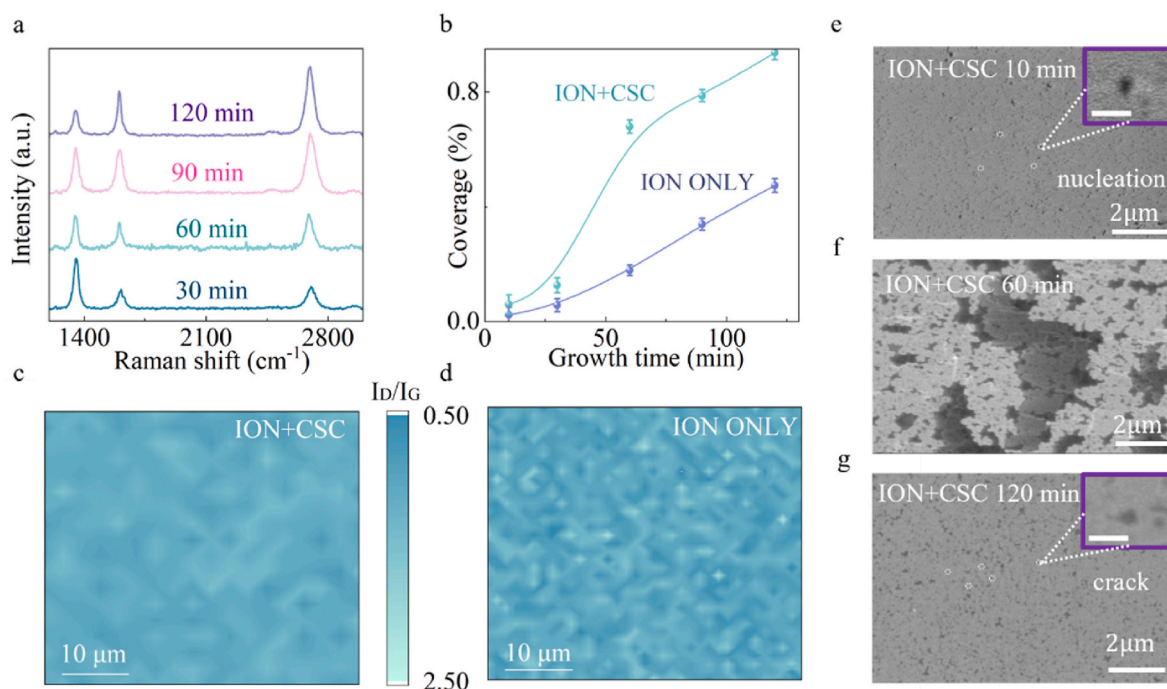


Fig. 2. Characterization of graphene. (a) Raman spectra of the quartz glass substrates after varying time from 30 min to 120 min graphene growth by APCVD at 1100 °C by the same flow rate of 2 sccm CH₄, 40 sccm H₂, 110 sccm Ar with the pattern of ION + CSC. (b) Coverage of graphene versus growth time with the pattern of ION + CSC and ION ONLY varying from 10 min to 120 min by APCVD at 1100 °C by the same flow rate of 2 sccm CH₄, 40 sccm H₂, 110 sccm Ar. (c–d) The value of I_D/I_G obtained through Raman mapping of graphene growth on quartz with ION + CSC and ION ONLY with the same growing condition of 2 sccm CH₄, 40 sccm H₂, 110 sccm Ar, 120 min growth time. (e–g) SEM images of graphene film growth assisted by Fe implantation at different stages for 10, 60, and 120 min on quartz glass with the type of ION + CSC (the scale bar in the illustration is 10 nm).

growth time reached 2 h, the value of I_{2D}/I_G of 1.80 and the Raman mapping ranging from 1.4 to 1.8 shown in Fig. S8a indicated that graphene was primarily monolayer [31], and the I_D/I_G ratio of 0.52 confirmed the high-quality graphene was fabricated.

Furthermore, we conducted a thoroughly comparison of the Raman bands of graphene grown in a confined environment with those grown in an unconfined environment with and without ion implantation. The results are presented in Fig. S2. Under the same growth conditions, the confined system exhibited significantly lower defect bands and higher SNR than the unconfined system, and the trend of the SNR with time confirms that the relative coverage content of graphene varied following different growth modes, the confined sample had the highest coverage ratio. As shown in Fig. 2c and d, both samples with ion implantations demonstrate that the spatially confined pattern exhibited higher uniformity, as evidenced by the mapping of I_D/I_G .

To gain an intuitive understanding of the graphene growth process, we characterized the morphology of graphene using Scanning Electron Microscopy (SEM) and quantified the coverage area of the samples to observe changes in coverage during growth. SEM is a technique used for high-resolution micro-area morphological analysis, allowing us to obtain morphological information about the samples directly with minimal damage and contamination. Fig. 2e, f, 2g, and Figure S5 indicate that, in samples with the same dose of catalyst nanoparticles, significant nucleation sites were found in the confined system after just 10 min growth, while the unconfined system only exhibited the same nucleation circumstance after 30 min. Additionally, we found that the confined system achieved full coverage in just about 2 h, whereas in our previous experiments, samples with ion implantation took about 5 h or longer to achieve full coverage. This indicates that the growth efficiency of the confined system is 2.5 times greater than that of the unconfined system. The morphology of graphene obtained through ion

implantation, including single-layer, double-layer, and multi-layer graphene, with single-layer graphene being the predominant form in the TEM images. Notably, no obvious metal nanoparticles were observed, which can be observed from Fig. S9 and further corroborated by the Raman mapping results, particularly the I_{2D}/I_G values, which ranging from 1.4 to 1.8.

To demonstrate that the catalysts exhibited a longer lifetime in the confined system, we conducted pseudo-in situ experiments to characterize the catalytic nanoparticles and quantitatively assess the relative content of the catalysts over time using X-ray Photoelectron Spectroscopy (XPS). The deconvolution results presented in Fig. 3a reveals two main peaks for Fe 2p_{1/2} and Fe 2p_{3/2}, confirming the successful implantation of catalyst ions. The results of curve fitting with Shirley background subtraction in the range of 695 eV–750 eV indicated the presence of Fe²⁺ from Fe₂SiO₄ at 724.29 eV, Fe at 705.8 eV, and Fe³⁺ at 703.4 eV [34], thus validating the ion implantation process. During the experimental process, specific time points were selected to cool down from the growth temperature directly. The XPS spectrum of the substrates at these time points were recorded to determine when the catalyst ions begin to become free of residues. This treatment was also applied to samples implanted without spatially confined ions, enabling a comparison of the amount of residual catalyst nanoparticles in both systems and verifying the role of ion confinement. The peak area of Fe 2p in the supplementary figures provide the relative content of Fe, and the results are presented in Figs. S3b and S3c and summarized in Fig. 3b. Fig. 3b illustrates the changes in the content of catalyst ions in spatially confined and unconfined systems over time, indicating that the gap indeed extended the lifetime of the catalyst ions. Our results indicate that in the spatially confined system, the duration of catalyst ions activity accounted for over 75% of the entire growth process, whereas in the unconfined system, the duration only accounted for 10%–20%, this

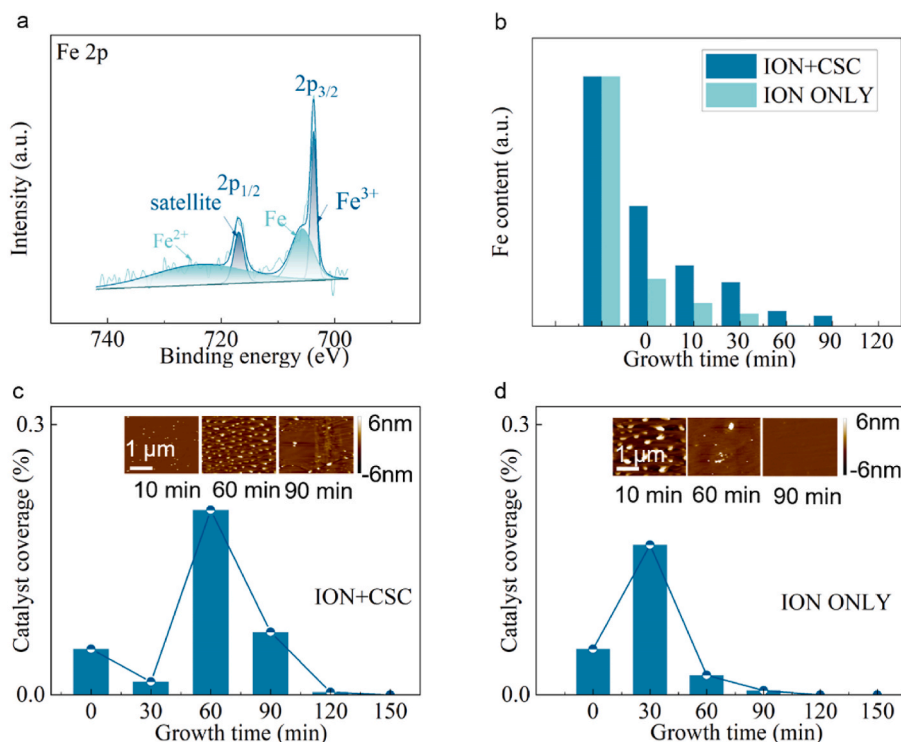


Fig. 3. Characterization of catalysts. (a) The deconvoluted Fe 2p spectrum derived from the ION ONLY pattern's XPS spectrum. (b) The Fe content of both the ION + CSC and ION ONLY patterns were statistically derived from the XPS spectrum, with growth times varying from blank (representing the quartz substrate without any growth process) via 0 min (representing the quartz substrate went through reduction process without growth process) to 150 min (representing the quartz substrate went through reduction process and growth process). This analysis was conducted using APCVD at a temperature of 1100 °C, maintaining a consistent flow rate of 2 sccm CH₄, 40 sccm H₂, 110 sccm Ar throughout the process. (c–d) The area of the entire substrate occupied by catalysts precipitation characterized by AFM with ION + CSC and ION ONLY patterns separately. (A colour version of this figure can be viewed online.)

represents a fourfold increase in catalytic activity.

To further illustrate the element distribution, we employed Time-of-flight secondary ion mass spectrometry (TOF-SIMS), featured by detecting elements at 10-ppb level and visualizing elements distribution through three-dimensional (3D) structural reconstruction at a high-resolution nanoscale, as shown in Fig. S10. Depth profile analysis of the ion-implanted quartz revealed that the signal intensity of Fe increased over time, peaking at approximately 50 s. By converting time into depth, the distribution of Fe ions was predominantly confined within the intended depth of ~20 nm within the quartz substrate, which was in consistent with Fig. 4a. The peak remained only as a weak signal after growth, which can be considered negligible.

Furthermore, to visualize this phenomenon, we characterized and quantified the morphology of the catalyst nanoparticles using Atomic Force Microscopy (AFM). To distinguish Fe ions from carbon species, we conducted pseudo-in situ experiments without carbon sources to observe the precipitation process of the catalysts, which were characterized using AFM, and the results are shown in Fig. 3c, d, and Figure S6. By analyzing the proportion of catalysts present on the substrate surface, we can determine the extent of the surface actively participating in the catalytic cracking of the carbon source. Our findings indicated that ions within the spatially confined gap exhibited a longer lifespan, allowing us to establish a strong correlation between catalyst distribution and growth efficiency.

To investigate the lifetime of catalyst ions in the confined system, we conducted a series of simulations and calculations from different perspectives to elucidate the mechanism of molecular motion within the gap. The outcomes are displayed in Fig. 4. The application of ion implantation technology in the field of two-dimensional material growth also relies on the tunability of its parameters. We utilized the mature software SRIM to simulate the range of ions with different energies within the substrate, providing theoretical guidance for the slow release of ions during our growth process. Fig. 4a illustrates the decomposition

of Fe ions within the quartz substrate at various extraction energies. By simulating the distribution of ions in the substrate, we can adjust different energy dosage parameters to modulate the distribution of catalyst nanoparticles. Additionally, in our system, the annealing process for quartz to repair caused the ions to reside close to the surface before growth, which simplified the subsequent simulations.

We subsequently employed the finite element method to simulate the airflow pattern within the gap [30], and the simulation details are provided in the supplementary information. Additionally, the results of the flow velocity along the middle lines with various gap distances are illustrated in Fig. 4b, which are picked up in Fig. S7. It indicated that if the airflow is primarily governed by fluid dynamics, with a gas flow velocity of 2 cm/s in the quartz tube, the flow velocity in the gap would be approximately 6×10^{-5} cm/s. This suggests that the time required for catalyst ions to traverse the gap would be on the order of 10^4 seconds, which contradicted with the experimental timescale we observed. In contrast, if molecular diffusion dominated the airflow pattern in the gap, the time for catalyst nanoparticles to cross the gap would be significantly shorter. Therefore, we concluded that the airflow pattern within the gap was predominantly controlled by molecular diffusion.

Based on molecular dynamics, the estimated time for nanoparticles to traverse the gap was about 3.34 s (details in supplementary information). Compared to experimental timescales, the migration of catalyst nanoparticles should be described by molecular diffusion rather than viscous dynamics. From this perspective, the relationship between the concentration of gaseous Fe nanoparticles (C_g), the survival time of catalyst ions (t), and the distance (d) could be expressed by the equation [32]:

$$C_g(d, t) = C_g(0, t) \exp\left(-\frac{d^2}{4Dt}\right) \quad (1)$$

In this equation, $C_g(d, t)$ and $C_g(0, t)$ represented the concentrations

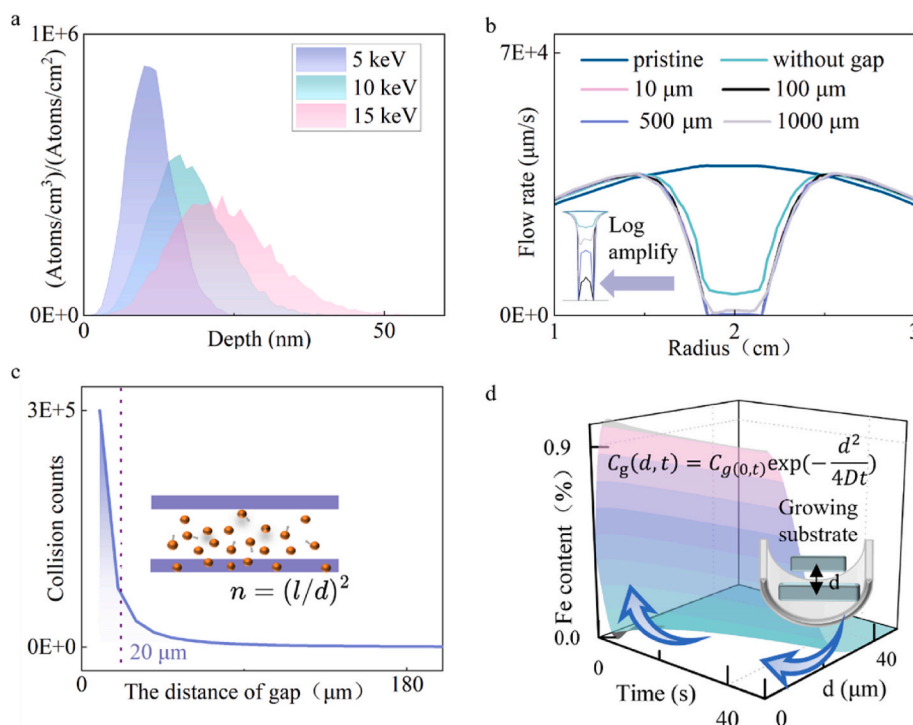


Fig. 4. Simulations of implantation and the airflow in the gap. (a) The range of Fe ions in a quartz substrate at 15 keV energy given by SRIM calculations. (b) The plots and the log amplify of the flow velocity profiles of the empty pipe, the original unrestricted pipe, and the restricted 10 μm, 100 μm, 500 μm, and 1000 μm gaps, respectively, using finite element simulation. (c) The average number for a nanoparticle colliding with the gap walls (n) in the process of traveling out of the gap, the illustration shows the schematic of the device for spatially confined growth in finite element simulation. (d) Distribution of the concentration of Fe ions in the system as a function of both the time of the process t and the distance of the gap d obtained by the method of Eq. (1). (A colour version of this figure can be viewed online.)

of gaseous Fe nanoparticles at distance d and the source substrate, respectively, with t as the time and D as the diffusion constant. The concentration of gaseous Fe nanoparticles varied significantly depending on both t and d , as illustrated in Fig. 4d, at the same time, the smaller distance d has the higher the concentration of catalyst ions. This framework allowed for the investigation of how Fe nanoparticle concentration influences controllable growth.

Additionally, due to the cage-like gap, the likelihood of collisions among carbon fragments, catalyst nanoparticles, and growth substrates was considerably enhanced, as described by the equation (please refer to the supplementary information):

$$n = \left(\frac{l}{d}\right)^2 \quad (2)$$

Here, n represents the average number of collisions among the entities traveling through the gap. This intuitive model is depicted in Fig. 4c, which aligned well with our experimental conclusions, explaining why the introduction of the gap significantly enhanced the catalytic cracking of carbon sources.

4. Conclusions

We have established a transfer-free CVD method that effectively combines ion implantation with catalyst spatial confinement, significantly enhancing graphene growth efficiency on insulating substrates. This novel approach reduces growth time on insulating materials, such as quartz glass, to approximately 2 h while extending the lifespan of catalyst ions, thereby facilitating precursor decomposition and preventing metal residue contamination—essential for semiconductor applications. The effectiveness of the spatially confined technique in improving the quality and characteristics of graphene underscores the critical interplay between physical geometry and material growth processes. Furthermore, this method holds promise for broader applications in the synthesis of other two-dimensional materials, for its revenue of thermodynamics and kinetic in catalytic reactions making it applicable to various other growth systems such as hexagonal boron nitride and molybdenum disulfide.

CRediT authorship contribution statement

Yang Xu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yue Li:** Writing – original draft, Formal analysis, Conceptualization. **Liu Qian:** Validation, Supervision. **Shurui Wang:** Methodology, Investigation. **Yunbiao Zhao:** Writing – review & editing, Validation, Supervision. **Ziqiang Zhao:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 52272033).

Y.X. and Y.L. prepared the samples and performed the experiments of synthesizing and characterizing graphene, assisted by L.Q. and S.W., Y. X. and Y.L. wrote and revised the manuscript with input from all authors. Y.Z. and Z.Z. supervised the overall projects. All authors contributed to the discussion and completion of this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbon.2025.120183>.

References

- [1] X. Xu, Z. Zhang, L. Qiu, J. Zhuang, L. Zhang, H. Wang, C. Liao, H. Song, R. Qiao, P. Gao, Z. Hu, L. Liao, Z. Liao, D. Yu, E. Wang, F. Ding, H. Peng, K. Liu, Ultrafast growth of single-crystal graphene assisted by a continuous oxygen supply, *Nat. Nanotechnol.* 11 (2016) 930–935, <https://doi.org/10.1038/nnano.2016.132>.
- [2] J.S. Bunch, S.S. Verbridge, J.S. Alden, A.M. van der Zande, J.M. Parpia, H. G. Craighead, P.L. McEuen, Impermeable atomic membranes from graphene sheets, *Nano Lett.* 8 (2008) 2458–2462, <https://doi.org/10.1021/nl801457b>.
- [3] D.C. Elias, R.R. Nair, T.M.G. Mohiuddin, S.V. Morozov, P. Blake, M.P. Halsall, A. C. Ferrari, D.W. Boukhvalov, M.I. Katsnelson, A.K. Geim, K.S. Novoselov, Control of Graphene's Properties by Reversible Hydrogenation: Evidence for Graphane 323 (2009) 610–613, <https://doi.org/10.1126/science.1167130>.
- [4] C. Lee, X. Wei, J.W. Kysar, J. Hone, Measurement of the elastic properties and intrinsic strength of monolayer graphene, *Science* (New York, N.Y.) 321 (2008) 385–388, <https://doi.org/10.1126/science.1157996>.
- [5] X. Wang, X. Li, L. Zhang, Y. Yoon, P.K. Weber, H. Wang, J. Guo, H. Dai, N-doping of graphene through electrothermal reactions with ammonia, *Science* (New York, N. Y.) 324 (2009) 768–771, <https://doi.org/10.1126/science.1170335>.
- [6] Y. Xie, T. Cheng, C. Liu, K. Chen, Y. Cheng, Z. Chen, L. Qiu, G. Cui, Y. Yu, L. Cui, M. Zhang, J. Zhang, F. Ding, K. Liu, Z. Liu, Ultrafast catalyst-free graphene growth on glass assisted by local fluorine supply, *ACS Nano* 13 (2019) 10272–10278, <https://doi.org/10.1021/acsnano.9b03596>.
- [7] R. Ye, J.M. Tour, Graphene at fifteen, *ACS Nano* 13 (2019) 10872–10878, <https://doi.org/10.1021/acsnano.9b06778>.
- [8] C.-Y. Su, A.-Y. Lu, C.-Y. Wu, Y.-T. Li, K.-K. Liu, W. Zhang, S.-Y. Lin, Z.-Y. Juang, Y.-L. Zhong, F.-R. Chen, L.-J. Li, Direct Formation of wafer scale graphene thin layers on insulating substrates by chemical vapor deposition, *Nano Lett.* 11 (2011) 3612–3616.
- [9] Y. Zhao, Y. Li, Y. Chen, Y. Chen, D. Zhou, Z. Zhao, Ion implantation assisted synthesis of graphene on various dielectric substrates, *Nano Res.* 14 (2021) 1280–1286, <https://doi.org/10.1007/s12274-020-3106-0>.
- [10] M.A. Fanton, J.A. Robinson, C. Puls, Y. Liu, M.J. Hollander, B.E. Weiland, M. LaBella, K. Trumbull, R. Kasarda, C. Howsare, J. Stitt, D.W. Snyder, Characterization of graphene films and transistors grown on sapphire by metal-free chemical vapor deposition, *ACS Nano* 5 (2011) 8062–8069, <https://doi.org/10.1021/nn202643t>.
- [11] J. Pang, R.G. Mendes, P.S. Wrobel, M.D. Wlodarski, H.Q. Ta, L. Zhao, L. Giebeler, B. Trzebicka, T. Gemming, L. Fu, Z. Liu, J. Eckert, A. Bachmatiuk, M.H. Rummeli, Self-terminating confinement approach for large-area uniform monolayer graphene directly over Si/SiO(x) by chemical vapor deposition, *ACS Nano* 11 (2017) 1946–1956, <https://doi.org/10.1021/acsnano.6b08069>.
- [12] G. Faggio, A. Capasso, G. Messina, S. Santangelo, T. Dikonimos, S. Gagliardi, R. Giorgi, V. Morandi, L. Ortolani, N. Lisi, High-temperature growth of graphene films on copper foils by ethanol chemical vapor deposition, *J. Phys. Chem. C* 117 (2013) 21569–21576, <https://doi.org/10.1021/jp407013y>.
- [13] S. Kasap, H. Khaksaran, S. Çelik, H. Özkaya, C. Yanik, I.I. Kaya, Controlled growth of large area multilayer graphene on copper by chemical vapour deposition, *Phys. Chem. Chem. Phys.* 17 (2015) 23081–23087, <https://doi.org/10.1039/C5CP01436K>.
- [14] L. Lin, B. Deng, J. Sun, H. Peng, Z. Liu, Bridging the gap between reality and ideal in chemical vapor deposition growth of graphene, *Chem. Rev.* 118 (2018) 9281–9343, <https://doi.org/10.1021/acs.chemrev.8b00325>.
- [15] X. Li, W. Cai, L. Colombo, R.S. Ruoff, Evolution of graphene growth on Ni and Cu by carbon isotope labeling, *Nano Lett.* 9 (2009) 4268–4272, <https://doi.org/10.1021/nl902515k>.
- [16] S. Park, R.S. Ruoff, Chemical methods for the production of graphenes, *Nat. Nanotechnol.* 4 (2009) 217–224, <https://doi.org/10.1038/nnano.2009.58>.
- [17] H. Kim, I. Song, C. Park, M. Son, M. Hong, Y. Kim, J.S. Kim, H.-J. Shin, J. Baik, H. C. Choi, Copper-vapor-assisted chemical vapor deposition for high-quality and metal-free single-layer graphene on amorphous SiO₂ substrate, *ACS Nano* 7 (2013) 6575–6582, <https://doi.org/10.1021/nn402847w>.
- [18] P.-Y. Teng, C.C. Lu, K. Akiyama-Hasegawa, Y.C. Lin, C.H. Yeh, K. Suenaga, P. W. Chiu, Remote catalyzation for direct formation of graphene layers on oxides, *Nano Lett.* 12 (2012) 1379–1384, <https://doi.org/10.1021/nl204024k>.
- [19] A. Ismach, C. Druzgalski, S. Penwell, A. Schwartzberg, M. Zheng, A. Javey, J. Bokor, Y. Zhang, Direct chemical vapor deposition of graphene on dielectric surfaces, *Nano Lett.* 10 (2010) 1542–1548, <https://doi.org/10.1021/nl9037714>.
- [20] J. Kwak, J.H. Chu, J.-K. Choi, S.-D. Park, H. Go, S.Y. Kim, K. Park, S.-D. Kim, Y.-W. Kim, E. Yoon, S. Kodambaka, S.-Y. Kwon, Near room-temperature synthesis of transfer-free graphene films, *Nat. Commun.* 3 (2012) 645, <https://doi.org/10.1038/ncomms1650>.
- [21] H. Kim, I. Song, C. Park, M. Son, M. Hong, Y. Kim, J.S. Kim, H.-J. Shin, J. Baik, H. C. Choi, Copper-vapor-assisted chemical vapor deposition for high-quality and metal-free single-layer graphene on amorphous SiO₂ substrate, *ACS Nano* 7 (2013) 6575–6582, <https://doi.org/10.1021/nn402847w>.
- [22] Z. Wang, Z. Xue, M. Zhang, Y. Wang, X. Xie, P.K. Chu, P. Zhou, Z. Di, X. Wang, Germanium-assisted direct growth of graphene on arbitrary dielectric substrates

- for heating devices, *Small* 13 (2017) 1700929, <https://doi.org/10.1002/smll.201700929>.
- [23] Y. Li, Y. Zhao, Y. Chen, Y. Chen, D. Zhou, Z. Zhao, Direct synthesis of single-layer graphene films on quartz substrate by a nanoparticle-assisted method, *Appl. Surf. Sci.* 529 (2020) 147082, <https://doi.org/10.1016/j.apsusc.2020.147082>.
- [24] X. Wang, W. Wan, S. Shen, H. Wu, H. Zhong, C. Jiang, F. Ren, Application of ion beam technology in (photo)electrocatalytic materials for renewable energy, *Appl. Phys. Rev.* 7 (2020) 041303, <https://doi.org/10.1063/5.0021322>.
- [25] T. Zhang, X. Wang, H. Liang, H. Zhang, G. Zhou, G. Sun, W. Zhao, J. Xue, Behaviour of MEVVA metal ion implantation for surface modification of materials, *Surf. Coating. Technol.* 83 (1996) 280–283, [https://doi.org/10.1016/0257-8972\(96\)02855-1](https://doi.org/10.1016/0257-8972(96)02855-1).
- [26] W. Huang, L. Gan, H. Yang, N. Zhou, R. Wang, W. Wu, H. Li, Y. Ma, H. Zeng, T. Zhai, Controlled synthesis of ultrathin 2D β -in2S3 with broadband photoresponse by chemical vapor deposition, *Adv. Funct. Mater.* 27 (2017) 1702448, <https://doi.org/10.1002/adfm.201702448>.
- [27] P.K. Mohapatra, S. Deb, B.P. Singh, P. Vasa, S. Dhar, Strictly monolayer large continuous MoS2 films on diverse substrates and their luminescence properties, *Appl. Phys. Lett.* 108 (2016) 042101, <https://doi.org/10.1063/1.4940751>.
- [28] Z. Tu, G. Li, X. Ni, L. Meng, S. Bai, X. Chen, J. Lou, Y. Qin, Synthesis of large monolayer single crystal MoS2 nanosheets with uniform size through a double-tube technology, *Appl. Phys. Lett.* 109 (2016) 223101, <https://doi.org/10.1063/1.4968582>.
- [29] S. Zhou, L. Gan, D. Wang, H. Li, T. Zhai, Space-confined vapor deposition synthesis of two dimensional materials, *Nano Res.* 11 (2018) 2909–2931, <https://doi.org/10.1007/s12274-017-1942-3>.
- [30] C.-C. Chen, C.-J. Kuo, C.-D. Liao, C.-F. Chang, C.-A. Tseng, C.-R. Liu, Y.-T. Chen, Growth of large-area graphene single crystals in confined reaction space with diffusion-driven chemical vapor deposition, *Chem. Mater.* 27 (2015) 6249–6258, <https://doi.org/10.1021/acs.chemmater.5b01430>.
- [31] C. Wang, W. Chen, C. Han, G. Wang, B. Tang, C. Tang, Y. Wang, W. Zou, W. Chen, X.-A. Zhang, S. Qin, S. Chang, L. Wang, Growth of millimeter-size single crystal graphene on Cu foils by circumfluence chemical vapor deposition, *Sci. Rep.* 4 (2014) 4537, <https://doi.org/10.1038/srep04537>.
- [32] Z. Lin, Y. Zhao, C. Zhou, R. Zhong, X. Wang, Y.H. Tsang, Y. Chai, Controllable growth of large-size crystalline MoS2 and resist-free transfer assisted with a Cu thin film, *Sci. Rep.* 5 (2015) 18596, <https://doi.org/10.1038/srep18596>.
- [33] A.C. Ferrari, D.M. Basko, Raman spectroscopy as a versatile tool for studying the properties of graphene, *Nat. Nanotechnol.* 8 (2013) 235–246, <https://doi.org/10.1038/nnano.2013.46>.
- [34] M.C. Biesinger, B.P. Payne, A.P. Grosvenor, L.W.M. Lau, A.R. Gerson, R.S.C. Smart, Resolving surface chemical states in XPS analysis of first row transition metals, oxides and hydroxides: Cr, Mn, Fe, Co and Ni, *Appl. Surf. Sci.* 257 (2011) 2717–2730, <https://doi.org/10.1016/j.apsusc.2010.10.051>.